Lecture 4 Exercise Solutions

1. Demonstrate the use of Diffie-Hellman Key Exchange in $Z_{9511}^{\times}$.

   9511 is prime, so $p = 9511$.

   <u>Alice</u>

   Given $p = 9511$, Alice needs to find a generator, g, for the entire group.

   $\varphi(9511) = 9511 - 1 = 9510 = 2 \cdot 3 \cdot 5 \cdot 317$

   Unfortunately, 9511 is not a Safe Prime, otherwise this would be easy. So, what are all the possible divisors of 9510? We know that there must be

   $$\binom{4}{1} + \binom{4}{2} + \binom{4}{3} + \binom{4}{4} =$$
   $$\frac{4!}{1!(4-1)!} + \frac{4!}{2!(4-2)!} + \frac{4!}{3!(4-3)!} + \frac{4!}{4!(4-4)!} =$$
   $$\frac{4!}{1!3!} + \frac{4!}{2!2!} + \frac{4!}{3!1!} + \frac{4!}{4!0!} =$$
   $$4 + \frac{4 \cdot 3}{2} + \frac{4 \cdot 3 \cdot 2}{3 \cdot 2} + 1 =$$
   $$4 + \frac{12}{2} + \frac{24}{6} + 1 =$$
   $$4 + 6 + 4 + 1 =$$
   $$15$$

   So, there are 15 possible orders of subgroups, since a subgroup order must divide the order of the group. One of these is the order of the group itself, so we only need to try 14, at most. Great!

   $$\binom{4}{1} = \frac{4!}{1!3!} = 4$$
   $$9510 / 2 = 4755$$
   $$9510 / 3 = 3170$$
   $$9510 / 5 = 1902$$
   $$9510 / 317 = 30$$

$$\binom{4}{2} = \frac{4!}{2!2!} = 6$$

$9510/2\cdot 3 = 9510/6 = 1585$

$9510/2\cdot 5 = 9510/10 = 951$

$9510/2\cdot 317 = 9510/634 = 15$

$9510/3\cdot 5 = 9510/15 = 634$

$9510/3\cdot 317 = 9510/951 = 10$

$9510/5\cdot 317 = 9510/1585 = 6$

$$\binom{4}{3} = \frac{4!}{3!1!} = 4$$

$9510/2\cdot 3\cdot 5 = 9510/30 = 317$

$9510/2\cdot 3\cdot 317 = 9510/1902 = 5$

$9510/2\cdot 5\cdot 317 = 9510/3170 = 3$

$9510/3\cdot 5\cdot 317 = 9510/4755 = 2$

Let's try $g = 2$. Detailed application of the Fast Exponentiation Algorithm will be omitted here. Let's try every possible exponent to see if the result is $\bar{1}$.

$2^{4755}\%9511 = 1$

We know 2 is not a generator in $Z^{\times}_{9511}$. Its order is either 4755 or some positive integer that divides 4755. We don't need to know. All we need to know is that it doesn't generate the entire group.

Let's try $g = 3$.

$3^{4755}\%9511 = 9510$

$3^{3170}\%9511 = 6020$

$3^{1902}\%9511 = 7104$

$3^{30}\%9511 = 2583$

$3^{1585}\%9511 = 6021$

$3^{951}\%9511 = 9124$

$3^{15}\%9511 = 6319$

$3^{634}\%9511 = 648$

$3^{10}\%9511 = 1983$

$3^{6}\%9511 = 729$

$3^{317}\%9511 = 4887$

$3^{5}\%9511 = 243$

$3^{3}\%9511 = 27$

$3^{2}\%9511 = 9$

Fantastic! So the subgroup generated by g is not a proper subgroup of $Z_{9511}^{\times}$. It generates the entire group. Let's test Euler's Theorem:

$3^{9510}\%9511 = 1$

Yeah, math works. Who would have thought?

OK, $p = 9511$ and $g = 3$. Now Alice needs to choose a random secret that she tells no one else: $a = 779$. Alice calculates $g^a \% p = 3^{779}\%9511 = 7711$.

Alice transmits the following to Bob:

$p = 9511$
$g = 3$
$g^a = 7711$

<u>Bob</u>

Bob receives

$p = 9511$
$g = 3$
$g^a = 7711$

Now Bob chooses a random secret that he tells no one else: $b = 317$. Bob calculates $g^b \% p = 3^{317} \% 9511 = 4887$. He also calculates $(g^a)^b \% p = 7711^{317} \% 9511 = 1989$. **1989 is the shared secret**.

Now Bob transmits the following to Alice:

$g^b = 4887$

Alice

Alice receives

$g^b = 4887$

Now Alice calculates $\left(g^b\right)^a \% p = 4887^{779} \% 9511 = 1989$ and **has 1989 as a shared secret with Bob that Eve cannot obtain**, absent some other form of attack like a Man-in-the-Middle.